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*Editorial*

# Multidisciplinary Design Optimization of Aircraft for Climate Neutral Aviation: Potential and Future Perspectives

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## 1. Introduction

The aviation sector has been forced to deal with a strong transformative phase driven by the urgent and non-negotiable goal of reducing its impact on the climate [1]. The continuous growth of the aviation industry and the associated increase in harmful anthropogenic emissions must be reduced as much as possible. Current forecasts predict a doubling of the number of passengers in the next 30 years [2], leading to a huge increase in greenhouse gas emissions if no actions are pursued [3–6]. Technological and methodological improvements are necessary to meet this ambitious target. On the one hand, novel technological solutions are necessary to improve the current state of the art, which is close to its maximum potential. On the other hand, each technological solution can not be investigated as an isolated techno-brick; but multidisciplinary and multi-fidelity approaches must be used in the aircraft design process in order to extract the maximum benefits of introducing new technologies. Novel technological solutions are currently being investigated in several research fields such as those concerning fuel, propulsion, and the airframe [7–10].

Regarding next-generation fuel and propulsion, three techno-bricks are currently being investigated, namely, hybrid-electric [11–13], hydrogen [14–16], and sustainable aviation fuels (SAFs) [17–19]. Hybrid-electric propulsion relies on the integration of thermal and electric power chains enabling fuel burn and emissions reduction [20–22], increase in propulsion efficiency [11], and noise mitigation [23]. Despite the numerous benefits, hybrid-electric propulsion systems face several critical barriers that currently limit their large-scale deployment. Technological barriers include limited power density and battery gravimetric energy density [11,24], thermal management challenges associated with high-power electric components [25,26], the electricity generation process [27] and lack of suitable infrastructures [28], and regulatory barriers [7]. Technological barriers limit the actual applicability of hybrid-electric propulsion to the regional sector, which occupies a small portion of the current aeronautical sector [29], thereby diminishing the impact of hybrid-electric propulsion on anthropogenic emissions. Furthermore, the production of electric energy plays a key role in the actual climate impact of hybrid-electric propulsion, and life cycle assessment is fundamental in this regard [30,31]. Regulatory barriers such as the lack of a certification process can delay the entry into service of this technology. Hydrogen represents a promising pathway toward climate-neutral aviation as it enables propulsion without $CO_2$ emissions and offers a gravimetric energy density three times higher than conventional kerosene. However, its very low volumetric energy density (four times lower than kerosene) poses major challenges for onboard storage. Liquid and cryo-compressed hydrogen are the most viable storage options, both requiring cryogenic conditions and complex thermal management necessary to maintain those conditions, boil-off losses, and, in the case of cryo-compressed tanks, heavier structures to withstand

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high pressure [32–34]. Beyond aircraft integration, the hydrogen production process and the lack of dedicated airport infrastructure represent major barriers to large-scale deployment [35,36]. SAFs are drop-in fuels designed to reduce greenhouse gas emissions over their entire life cycle without requiring major changes to aircraft or infrastructure. Produced from waste-based, non-food, or synthetic feedstocks, SAFs can significantly lower life-cycle $CO_2$ emissions [37]. However, current certification limits blending with conventional kerosene to 50%, constraining in-flight emission reductions [17]. Major barriers to large-scale adoption include high production costs, limited market penetration, feedstock availability, and insufficient regulatory support [38]. While bio-based SAFs face scalability concerns, power-to-liquid pathways offer a promising alternative, albeit at a substantial cost, and life cycle $CO_2$ emission reduction if produced according to renewable electricity requirements [18].

Regarding the airframe, two paths are currently under investigation: (*i*) unconventional airframe designs to enhance aerodynamic efficiency, reduce fuel consumption, and lower emissions [39,40]; and (*ii*) advanced aero-propulsive integration [41]. Unconventional airframe concepts, such as box-wing [42], blended-wing-body [43], and truss-braced [44], are particularly promising for their potential lift-to-drag improvements [45–47], enhanced structural efficiency [48–50], and advanced integration of novel propulsion systems, such as hybrid-electric [51–53] and hydrogen [54–56] systems. Advanced aero-propulsive integration is the study of the interaction between the propulsion system and the aircraft airframe, aiming to enhance the aircraft's aerodynamic performance. Currently, investigated topics are distributed electric propulsion [57], boundary layer ingestion (BLI) [58], and next-generation engines, such as ultra-high-bypass ratio [59] and open-rotor [60]. Distributed electric propulsion can enhance high-lift capability and potentially improve cruise efficiency by reducing wing wetted surface. However, its implementation is limited by increased structural complexity, weight penalties due to the presence of multiple motors along the wing, and demanding thermal management requirements [25,61]. BLI improves overall aircraft efficiency by re-energizing the low-momentum boundary layer, thereby reducing wake drag and jet mixing losses: preliminary studies indicate potential block-fuel reductions of 5–12% [62]. However, BLI still faces key challenges, including flow distortion management and complex structural integration [58]. Ultra-high bypass ratio turbofan and open rotors increase propulsive efficiency, but their integration poses significant challenges such as aerodynamic interference with the airframe, ground clearance, or noise (particularly in the case of the open rotors).

The above paragraphs have highlighted that numerous technological solutions are currently being investigated, demanding the development of proper design tools to assess the actual impact at the aircraft level. The first key enabler is the adoption of a multi-fidelity approach where simulation techniques, ranging from low- to high-fidelity, allow for the accurate investigation of aerodynamics, structures, propulsion integration, and mission performance in early-design phases, thus reducing uncertainty and design risk [63–65]. Building on this, multidisciplinary design optimization (MDO) represents a second strategic pillar, enabling the integrated treatment of aerodynamics, structures, propulsion, cost, emissions, and operational aspects within unified optimization environments. These frameworks support design-space exploration, trade-off identification, the high-fidelity integration of design tools and treatment of uncertainties [66–68]. A third methodological innovation is the implementation of digital twins (DTs), i.e., dynamic virtual replicas of aircraft and subsystems continuously updated through sensor data and physics-based models. DTs enhance lifecycle management, predictive maintenance, and structural health monitoring [69–71]. Finally, a fourth emerging pillar is artificial intelligence, used to accelerate aerodynamic and structural optimization, create surrogate models, analyze aeroelastic

phenomena, support uncertainty quantification, and enable damage detection, especially when combined with physics-informed approaches to ensure physical reliability [72–75].

As described above, the aviation sector is evolving to target the sustainability goal. Regarding propulsion, hybrid-electric-, hydrogen-, and SAF-powered aircraft are the most promising solutions, but due to their various weaknesses, SAF is expected to be the most practical near-term solution, hydrogen is regarded as a long-term decarbonization option, and batteries are regarded as a regional air mobility solution [76,77]. Unconventional airframes and advanced aero-propulsive integration are promising solutions, particularly when coupled with novel propulsion technologies; however, an increase in knowledge and maturity is to be pursued in order to increase the technology readiness level. In this regard, the development of advanced interdisciplinary design tools play a crucial role in designing robust, efficient, and sustainable next-generation aircraft. Within this broader framework, shaped by the evolution of propulsion technologies, airframe solutions, and innovative design methodologies, as well as by the need to assess their future potential and integration, this first edition of the Special Issue "Multidisciplinary Design Optimization of Aircraft for Climate-Neutral Transport Aviation", available here [78], was conceived. This Special Issue set out to collect the most up-to-date methods, modeling strategies, and research results for the design of next-generation aircraft. In what follows, an overview of the research works published in this Special Issue is provided.

## 2. Overview of the Published Articles

A total of nine contributions are included in this Special Issue, addressing the topic of the multidisciplinary design optimization of aircraft to mitigate their impact on the climate. The paragraphs below provide an overview of these works, outlining their scope and emphasizing the key results achieved by each study.

Ref. [79] proposes an integrated operational and fleet-level methodology to assess the transition of short- and medium-haul aviation from conventional jet fuel to hydrogen propulsion under realistic infrastructural constraints. The approach combines preliminary aircraft sizing, route operability analysis, and fleet assignment optimization, explicitly accounting for the uneven availability and cost of hydrogen refueling across airports. A key innovation is the systematic modeling of hydrogen tinkering (i.e., carrying additional hydrogen from the departure airport in order to enable the return flight or reduce refueling at the destination) as an enabler of network coverage, allowing hydrogen-powered aircraft to reach destinations without refueling infrastructure within specific range limits, while quantifying the associated energy and cost penalties. The methodology is applied to representative European networks, exploring the impact of aircraft design range reduction on route coverage and efficiency. Results indicate that depending on hydrogen infrastructure availability and aircraft design range, hydrogen-powered aircraft can operate between 81% and 96% of the short- and medium-haul network considered. Building on this operational feasibility, the optimized fleet transition scenarios show that the associated aviation-induced temperature change can be reduced by up to 57% relative to an all-jet-fuel baseline, as quantified using the Average Temperature Response metric.

Ref. [80] presents an efficient methodology for mission power management optimization of parallel hybrid-electric regional aircraft suitable for the conceptual design phase. The methodology is embedded in the THEA-CODE multidisciplinary design framework and focuses on minimizing fuel consumption through the optimization of the power supply profile. In particular, the thermal engine supplied power is modeled as a time piecewise function composed of a pre-defined number of intervals: the value of thermal power supplied in each interval is treated as design variable, whereas the electric contribution is derived from the aircraft flight equilibrium equations. The main innovation lies in the

parametric study on the number of intervals composing the piecewise function describing the power supplied by the thermal engine. The approach is applied to a 40-passenger regional aircraft with an entry into service in 2035 and design range of 600 nm. The results show that the optimum thermal power supply profile is composed of three intervals since finer discretizations do not yield additional performance benefits. The optimized configurations achieve significant block fuel reductions compared to a full-thermal reference at the expense of increased battery mass and MTOW.

Ref. [81] investigates the integration of an auxiliary solar power system (SPS) for short-range, daylight-operated aircraft, using seaplane air taxi operations as a representative case study. An enhanced multidisciplinary design and analysis framework is developed to assess SPS feasibility at the aircraft level, explicitly accounting for mission-dependent effects such as solar irradiance variability, cloud coverage, temperature-dependent photovoltaic efficiency, and realistic system weight estimation, including encapsulation and wiring. The framework is embedded within a multidisciplinary analysis and optimization environment and applied to Harbour Air's DHC-2 Beaver operations. Results show that during peak seasonal operations, the SPS can supply the aircraft's secondary electrical loads for up to 86% of mission time and generate excess power that could supplement hybrid-electric propulsion. Although current technologies yield modest fuel-burn reductions at mission level, a design-of-experiments analysis highlights clear trends toward increasing benefits, with higher cell efficiency and lower system mass. This study demonstrates that solar-assisted architectures can contribute to emissions reduction and improved ground operations, supporting their role as a viable technology enabler for climate-neutral regional aviation.

Ref. [82] presents a framework to quantify the full climate impact of individual real-world flights by explicitly accounting for both $CO_2$ and non-$CO_2$ effects. The methodology combines validated aircraft and engine models with atmospheric conditions that vary along the flight trajectory, allowing for the simulation of realistic operating conditions. Emissions of $NO_x$ and soot are evaluated using multiple correlation methods, while contrail effects are evaluated by explicitly accounting for the formation of persistent contrails along the flight trajectory. The computed emissions are then used as inputs to a Linear Temperature Response model, which estimates the resulting change in global surface temperature over a 77-year time horizon. The results show that non-$CO_2$ effects driven by NOx emissions dominate the climate response shortly after the flight, whereas $CO_2$ becomes the main contributor over longer time horizons. Moreover, for short-range missions, the climb phase is identified as a relevant contributor to the overall climate impact, highlighting the importance of segment-level analyses. Sensitivity analyses reveal differences up to 13–14% in temperature response depending on the emission modeling strategy, underscoring the importance of integrated, flight-specific climate assessment tools.

Ref. [83] presents a conceptual-level assessment of a radical short–medium-range aircraft configuration based on a blended-wing-body (BWB) airframe combined with turbo-electric distributed propulsion, developed within the European IMOTHEP project. The methodology relies on an integrated aircraft–mission–powertrain modeling framework that enables consistent whole-aircraft performance assessments under representative top-level aircraft requirements and 2035 entry-into-service technology assumptions. The main innovation lies in the quantitative isolation of the propulsion system contribution from the intrinsic aerodynamic benefits of the BWB architecture, accounting for both boundary-layer ingestion effects and the mass and drag penalties associated with electric components. Results indicate that the turbo-electric BWB configuration achieves a measurable reduction in fuel consumption compared to an A320neo-based reference aircraft. However, when benchmarked against an advanced BWB configuration with conventional turbofan propul-

sion and comparable technology assumptions, the additional fuel-saving potential of the turbo-electric architecture is found to be limited.

Ref. [84] investigates the potential of advanced pilot assistance systems to simultaneously reduce fuel consumption and noise during aircraft approach. Building on the Low-Noise Augmentation System (LNAS), a pilot assistance concept that continuously monitors the aircraft energy state to support energy-optimal, low-noise descents, the study introduces enhanced Flight Management System (FMS) functionalities developed within the SESAR DYNCAT project, aimed at improving aircraft energy management under realistic air traffic constraints. The methodology relies on real-time pilot-in-the-loop cockpit simulations using an Airbus A321 FMS test bench, coupled with high-fidelity fuel consumption assessment and detailed noise modeling via sonAIR, a validated high-fidelity aircraft noise simulation framework that models both airframe and engine noise and their propagation to the ground. Results show that the proposed system enables more uniform and optimized approach profiles, with engines operating mostly at idle; later deployment of high-lift devices and landing gear; and reduced use of speed brakes. On average, fuel savings of about 5 kg per approach are achieved, alongside localized noise reductions in the critical final approach segment. The study also highlights the existence of Pareto-optimal trade-offs between noise and fuel, demonstrating that operational optimization requires multi-objective strategies rather than single-metric minimization.

Ref. [85] presents a preliminary performance assessment of a box-wing medium-range aircraft powered by liquid hydrogen ($LH_2$). An optimization-based multidisciplinary design framework is developed to retrofit the kerosene-based box-wing aircraft designed in the PARSIFAL project by integrating cryogenic hydrogen tanks within the fuselage. The core methodological contribution lies in the systematic exploration of tank sizing and layout solutions, accounting for thermo-structural tank design, hydrogen thermodynamics, and mission-level performance simulation. The study highlights the key trade-offs between available internal volume for $LH_2$ storage, payload capacity, and achievable range. Results show that liquid hydrogen integration leads to significant payload penalties compared to the kerosene-fueled baseline, with reductions of about 50% at design range (5700 km) and around 25% for short-range missions (1500 km). Despite these penalties, the box-wing architecture emerges as a key enabler for hydrogen-powered operations. The retrofitted configuration can still transport approximately 230 passengers over 1500 km or about 170 passengers over 4800 km, figures that remain compatible with current medium-haul requirements. By contrast, an equivalent hydrogen retrofit applied to a conventional tube-and-wing aircraft can transport 90 passengers for a 4800 km mission.

Ref. [86] proposes a data-driven competency assessment framework for civil aviation pilots operating under wind shear conditions, addressing a critical safety challenge linked to adverse meteorological events. The study develops a wind-shear-operation-based competency model by decomposing pilot actions into observable behaviors and task-specific check items aligned with ICAO and IATA competency frameworks. Methodologically, the authors integrate three-dimensional competency assessment, capturing behavioral frequency, behavioral coverage, and threat-and-error management outcomes, with an optimization algorithm that automatically calibrates quantitative evaluation thresholds from historical simulator training data. The principal innovation lies in transforming traditionally subjective examiner judgments into a transparent and reproducible assessment process grounded in operational data. Validation using simulated wind shear scenarios from training session of 200 pilots demonstrates strong agreement with expert evaluations, achieving a correlation coefficient of 0.854 and an overall assessment accuracy of 93.33%. The proposed framework enhances the consistency, interpretability, and scalability of pilot

competency evaluation, offering a robust foundation for evidence-based training under complex and safety-critical flight conditions.

Ref. [87] proposes a gate-to-gate simulation environment designed to capture both air traffic dynamics and localized operational details within the European Air Traffic System. The methodology combines flight trajectories and ground operations: airborne operations are modeled through BADA (Base of Aircraft Data), which provides a specific trajectory according to a predefined route, ground operations are instead described by a discrete-event simulation that explicitly represents turnaround processes and selected taxiing activities at airports. This allows for the analysis of interactions between airports, terminal areas, and traffic flows, allowing local operational phenomena to be studied in terms of their impact on the overall air traffic network. Monte Carlo simulations, driven by real operational delay statistics, demonstrate the capability of the framework to quantify network-scale performance effects such as delay propagation. The results highlight the suitability of the proposed environment as an facilitating tool for assessing future operational concepts, including the large-scale introduction of alternative-propulsion-based aircraft.

## 3. Concluding Remarks

Climate-neutral aviation cannot rely on isolated technological breakthroughs alone. Novel propulsion solutions such as hybrid-electric, hydrogen, and sustainable aviation fuels, together with unconventional airframes and advanced aero-propulsive integration concepts, require advanced modeling, simulation, and optimization environments. Multi-fidelity and multidisciplinary optimization frameworks, digital twins, and AI-driven tools emerge as key methodological enablers capable of consistently integrating aerodynamics, structures, propulsion, operations, and climate metrics from the early-design stages. Only through this tight coupling of technology development and robust modeling strategies can the aviation community assess realistic impacts, reveal trade-offs, manage uncertainties, and effectively support the transition toward sustainable aircraft concepts.

The contributions included in this Special Issue have strengthened this position, addressing hydrogen transition strategies at fleet and network level, hybrid-electric power-management optimization, the integration of solar-powered aircraft to support regional operations, the quantification of climate impact, including non-$CO_2$ effects, radical configurations such as blended-wing-body with turbo-electric propulsion, and hydrogen integration within box-wing architecture. Complementary research has highlighted the role of advanced pilot assistance systems in simultaneously mitigating noise and fuel consumption, proposed data-driven pilot competency assessment frameworks for safety-critical operations, and introduced gate-to-gate simulation tools to understand future traffic and operational scenarios. Collectively, these papers demonstrate the central role of multidisciplinary, multi-scale, and multi-objective design environments in assessing the feasibility, performance, and climate effectiveness of next-generation aircraft.

The results presented in this first edition clearly underline both the maturity of the field and the significant work still required to achieve sustainability goals. Building on the scientific advances gathered herein, the next edition of this Special Issue [88] will further expand in scope, encompassing higher technology readiness, enhanced integration on technological innovations and modeling frameworks, and stronger links with operations, infrastructure, and regulatory evolution. The aim is to continue fostering a comprehensive, multidisciplinary, and forward-looking discussion, providing the community with the robust tools and insights necessary to design, enable, and evaluate the sustainable aircraft and air transport systems of the future.

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